VoteLens: The Courage to Be Wrong

Evidence Traceability and Failure Anatomy in Multilingual Document-Grounded NLI over Swiss Voting Booklets

Track 2A, OST • 15 October 2026 • Exploratory development evidence

Abstract

We present VoteLens, a diagnostic system for document-grounded natural-language inference over official Swiss voting booklets. Given a booklet and a claim, the system retrieves page-grounded evidence, classifies entailment, neutral or contradiction using Apertus v1.5, and validates the model answer against an output contract (vote identity, citation occurrence, schema). Three observations drive this report; all are single-run analyses on *exposed development data*, not held-out benchmarks.

(1) Headline — coverage ≠ correctness. A strict `vote-aware-v1` candidate achieved slightly higher raw-label accuracy (41/54 vs 40/54) yet dramatically lower valid-output coverage (25/54 vs 43/54). Of its 29 rejections, 18 were gold-correct; among 18 gold-neutral cases, none became a valid accepted output. Conditional accuracy among accepted answers was 23/25 = 92% while overall strict correctness was 23/54 = 42.6% (paired exact McNemar $p = 0.122$): *an apparently reliable system can look better when evaluated only on accepted answers, while rejecting many otherwise correct ones.* The 54-case comparison uses Apertus v1.5 70B, the selected 70B submission model, and changes several pipeline components together — it is not a single-factor causal experiment.

(2) Retrieval scope. Focusing on five within-booklet passages beat full-booklet prompting (accuracy 0.802 vs 0.519, $n = 81$ paired cases) at 96.3% fewer input tokens. On a separate frozen 81-case development sample, optimising a lexical reference-recall proxy did *not* improve downstream NLI: dense retrieval scored 0.827 (macro-F1 0.824) but proposition reranking raised Recall@5 from 0.580 to 0.716 while accuracy fell to 0.790.

(3) Placement observation. In a 15-case diagnostic, moving the same decisive evidence behind distractors — identical input-token counts — produced three label regressions. These are single-run observations; they do not establish a positional cause.

All three are exploratory development evidence. No causal attribution follows from these data.

1. Introduction

Swiss direct democracy places exceptional demands on evidence reliability. Citizens must adjudicate complex referenda using official booklets that are multilingual (DE/FR/IT), structurally repetitive, and frequently place supporting and opposing arguments in the same document. Martin et al. (2024) identify precisely this setting as a high-stakes domain for LLM-based retrieval-augmented generation and frame the civic risk as potential *erosion of shared epistemic ground* upon which democratic deliberation depends. Their position paper motivates semantic verification frameworks that validate the factual content of LLM-generated summaries before they reach voters, proposing knowledge-graph-assisted cross-checking as a possible architecture.

VoteLens addresses a narrower but adjacent question: given that a judge model *has* produced a label and a cited passage, can we trace whether the cited evidence textually occurs in the supplied booklet (mechanical provenance), and what failure modes arise at the interface between retrieval and judgement? We do not implement the knowledge-graph verification layer of Martin et al. (2024); our development experiments expose a concrete gap between output-contract validation (citation string matching) and semantic evidence sufficiency (whether a passage *establishes* a claim), which their framework would need to resolve before deployment.

Because booklets are per-language editions of one vote day, the case analyses in §3 suggest distinct proposal-scope and stakeholder-attribution failure modes rather than simple topic-matching errors. These are post-hoc attributions from paired error inspection, not statistical clustering.

The submission CLI accepts a specified booklet and claim, returns entailment/neutral/contradiction with page-grounded citations. Retrieval chunks the supplied booklet and retrieves Top-5 passages using SentenceSwissBERT mean-pooled cosine similarity (Vamvas et al., 2023). Gold references and labels never enter extraction, ranking or judgement.

Three open research questions organise the attribution analysis:

RQ1. Can a model distinguish uncertainty about a referendum's identity from insufficient evidence for a claim?

RQ2. Does independent vote grounding reduce invalid outputs without sacrificing evidence provenance?

RQ3. How much of the observed failure is attributable to retrieval scope, NLI reasoning, and output-contract compliance?

These are questions for future controlled tests, not established causal findings.

2. Method

2.1. Pipeline

The application retrieves five 900-character passages from the specified booklet. The judge receives the claim, retrieved evidence and speaker-attribution instructions, and must output a JSON object with a label code (0 = entailment, 1 = neutral, 2 = contradiction), cited passage IDs and source spans. The application validates vote identity, page-relative offsets and citation occurrence before rendering the official response. A citation is *verified* only against supplied evidence; verification establishes textual provenance, not semantic support.

The submission target is `swiss-ai/Apertus-v1.5-70B` via CSCS. Local experiments use the third-party conversion
`m1rkocasu/Apertus-v1.5-8B-text-MLX-6bit` (revision 9a3c32df, 6-bit quantisation). The 54-case paired comparison uses `swiss-ai/Apertus-v1.5-70B` via CSCS; its results do not establish official 8B performance.

2.2. Experiment design

Study	n	Model	Purpose
Vote-aware paired	54	70B CSCS	Complete-system comparison
Full-booklet vs retrieval	81	8B MLX	Evidence scope
Oracle diagnostic	19	8B MLX	Judgement under gold evidence
Retrieval ablation (3 arms)	81	8B MLX	Ranker effect on NLI
Evidence sensitivity (4 cond.)	15	8B MLX	Placement and sufficiency

All experiments used temperature 0, 512-token output, strict JSON parsing; failed/invalid outputs remain in denominators.

An identity-aware prompt improved Macro-F1 from 0.656 to 0.850 on nine targeted cases, but yielded no improvement on a separate 20-case development sample (0.633 for both variants); we therefore retained the original prompt. These are local development results, not official evaluation scores or evidence of generalisation.

Metric definitions. Four distinct measures: *raw-label accuracy* (model label matches gold, contract irrelevant; some correct raw labels failed citation verification but count here); *strict correctness* (correct label AND accepted response); *valid-output coverage* (fraction passing the full output contract); *citation verified* (mechanical source-string occurrence, not semantic support). The 15-case sensitivity diagnostic has only entailment and contradiction gold labels; its macro-F1 averages over two classes and is not directly comparable with three-class F1. Deterministic JSON recovery (stripping fences, extracting the first JSON object) is applied before the label parser; this is format normalisation, not semantic interpretation.

2.3. Failure taxonomy

Failures are assigned to the earliest independently measurable stage:

L1 Retrieval scope — evidence left the named vote, or the decisive proposition was displaced by the ranker.
L2 Semantic sufficiency — evidence is relevant but does not *establish* the claim. *No independent annotation of this layer exists yet.*
L3 Speaker attribution — the right proposition, credited to the wrong stakeholder.
L4 Output contract — a correct raw label rejected by the identity/citation gate.

Two distinctions maintained throughout: **output-validation failure ≠ reasoning failure**, and **evidence supported ≠ evidence sufficient**.

3. Results

Separate production-path development checks (not the paired comparison). On local, exposed 60-case CSCS Apertus v1.5 70B evaluations, Task A (`vote-aware-v1`) achieved three-class Macro-F1 **0.6635**, with **46/60** valid outputs (14 rejected), against the Task A qualification threshold of 0.60. Task B achieved Macro-F1 **0.8418**, with **60/60** valid outputs, against its 0.75 threshold. These are neither official nor held-out results and do not establish qualification. The 54-case comparison below uses a different development sample; its `vote-aware-v1` Macro-F1 of 0.491 must not be conflated with this run.

3.1. Headline: paired vote-aware comparison (the coverage result)

54 paired cases on CSCS Apertus v1.5 70B. Two examples per language-pair × gold-label cell, alternating AB/BA order, 108 accounted calls. This is a complete-system comparison; it cannot isolate the effect of vote awareness.

Metric	`dense-v1`	`vote-aware-v1`
Raw label accuracy	40/54 (0.741)	41/54 (0.759)
Strict accuracy	32/54 (0.593)	23/54 (0.426)
Valid-output coverage	43/54 (0.796)	25/54 (0.463)
Rejections	11	29
Macro-F1 (invalid = error)	0.652	0.491

The candidate's 29 rejections: 14 `vote_unrecognised`, 15 `invalid_vote_status`. All 29 still contained a parseable raw NLI label; **18 matched gold**. Among the 18 gold-neutral cases, 12 returned a correctly neutral raw label; none became an accepted output.

On strict correctness: 18 dense-only correct, 9 candidate-only correct, 14 both correct, 13 both incorrect. Exact McNemar $p = 0.122$.

Denominators and metric selection. The `vote-aware-v1` candidate's accepted-output accuracy is 23/25 = 92% (conditional on passing the gate), while its strict correctness over all 54 cases is 23/54 = 42.6%. The `dense-v1` conditional accuracy is 32/43 = 74.4%. Reporting only the conditional figure would create a misleading impression. Hutiri et al. (2024) demonstrate empirically — in the unrelated domain of speaker verification bias — that base-metric selection, the choice between ratio and difference measures, and aggregation methodology can produce contradictory conclusions from identical raw data. We adopt the same caution: no single metric should be quoted as *the* system accuracy. The paired comparison changes multiple factors simultaneously; it is not a single-factor causal experiment.

The immediate rejection mechanism is the identity gate. However, *where* an output was rejected is distinct from *why* the answer went wrong. In row 728, all five retrieved passages concerned the Climate and Innovation Act for a COVID-19 claim — a retrieval-scope failure alongside an identity failure. In row 1115, the evidence explicitly named the requested initiative and the raw label was correct, yet the gate rejected the output. In row 964, several passages identified the cash initiative, yet the model declared it unrecognised and produced an incorrect label. These three cases are not one mechanism; the identity gate alone is not claimed to have caused every observed error.

This is not selective classification (Geifman and El-Yaniv, 2017): the gate does not optimise a coverage–risk frontier.

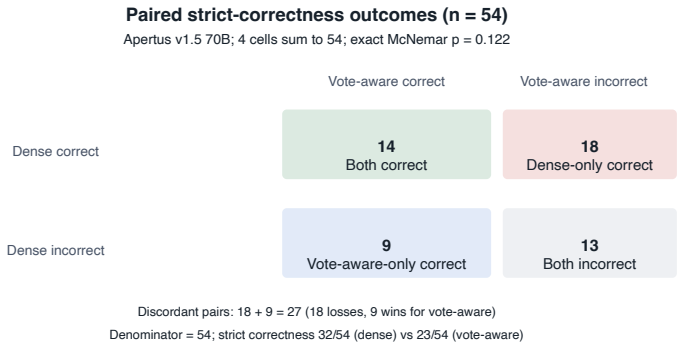


Figure 1: Paired strict-correctness outcomes for the 54-case vote-aware comparison (Apertus v1.5 70B) — the central figure. Four cells: both correct 14, dense-only correct 18, vote-aware-only correct 9, both incorrect 13. Exact McNemar $p = 0.122$.

3.2. Supporting evidence: retrieval scope and the retrieval ablation

Full-booklet vs Top-5 retrieval on 81 paired development cases (8B MLX): accuracy 0.519 → 0.802 at 96.3% fewer input tokens; one retrieved run hit the output limit (80/81 completed). This motivated the retrieved-evidence architecture; it is a development design decision, not a benchmark result.

Three retrieval arms compared on a separate frozen 81-case sample (27 per NLI label, 9 per DE/FR/IT combination). Method A: dense SentenceSwissBERT. B: dense + BM25 with RRF (Cormack et al., 2009). C: B + proposition reranking. All 243 calls completed without errors.

Method	Acc	Macro-F1	R@5	Cit. verif.	Tokens
A • dense	0.827	0.824	0.580	51.9%	1,215
B • hybrid RRF	0.802	0.797	0.617	49.4%	1,326
C • prop. rerank	0.790	0.784	0.716	44.4%	1,314

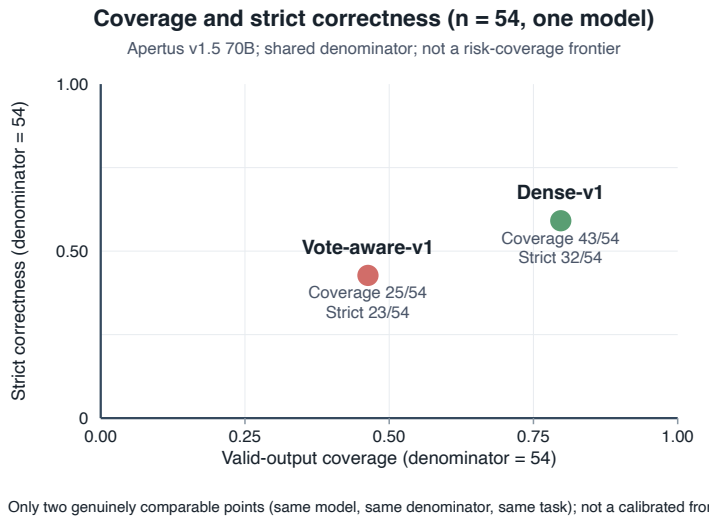


Figure 2: Valid-output coverage versus strict correctness for the 54-case paired comparison (one model, one denominator, shared axis transform). Two genuinely comparable points. Not a risk-coverage frontier; the gate was not calibrated and no direction of improvement is implied.

Key observation: C increased R@5 by 11/81 cases yet produced three fewer correct labels. Paired sign tests on discordant pairs (8–11 per transition) do not reach significance.

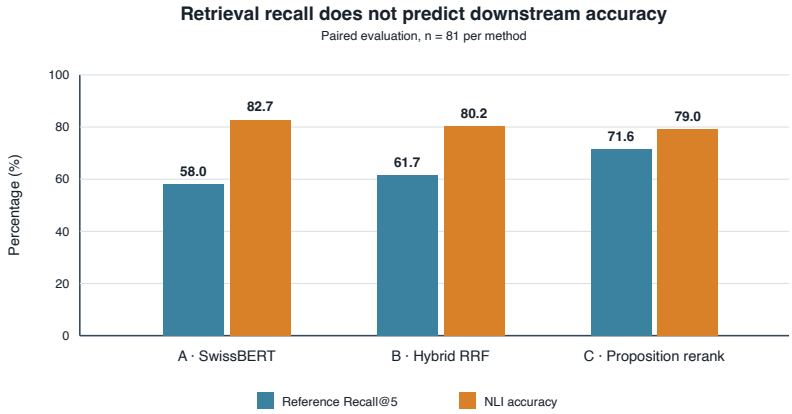


Figure 3: Reference Recall@5 versus NLI accuracy (81-case ablation). Source: `data/experiments/retrieval-ablation-v1/summary.json`.

3.3. Supporting evidence: evidence sensitivity diagnostic (exploratory)

15 post-hoc selected cases (8 entailment, 7 contradiction; no neutral gold), 60 calls, 8B MLX. Four conditions per case: A (original Top-5), B (manually constructed sufficient evidence), C (B + two distractors), D (identical to C, B block moved *behind* distractors, identical token counts to C).

Condition	Correct	Acc	Macro-F1	Input tokens
A • original Top-5	8/15	0.533	0.593	18,837
B • minimal sufficient	15/15	1.000	1.000	5,784
C • B + distractors	14/15	0.933	0.933	9,570
D • distractors + B	11/15	0.733	0.786	9,570

C→D produced three regressions (rows 1026, 1344, 189) with identical passage multisets and identical per-case token counts. This relates to positional sensitivity in long contexts (Liu et al., 2024), but cannot establish the same mechanism from 15 cases. Row 1344: the model correctly states "4%, not 10%" yet labels the claim neutral (gold: contradiction) — an output inconsistency, not a parser error.

B represents a case-informed upper bound; only 11/15 B citations passed the strict source-occurrence check.

4. Positioning and Related Work

Civic information infrastructure. Martin et al. (2024) provide the closest conceptual antecedent. Their position paper motivates semantic verification of LLM-RAG outputs in Swiss direct democracy, proposes a knowledge-graph framework for cross-referencing generated claims against structured sources, and identifies the civic risk of unverified generation during referenda. VoteLens does not implement their knowledge-graph verification; our contribution is empirical rather than architectural: we expose what happens at the *evidence-to-label* interface when provenance checking (string occurrence) is present but semantic sufficiency adjudication is absent, and show that the resulting coverage failure is invisible to any metric that counts only accepted outputs.

Evidence-grounded NLI and retrieval. The retrieve-then-judge pipeline structure is established (Lewis et al., 2020; Thorne et al., 2018). Our contribution is not the architecture but the failure anatomy: showing that in Swiss booklets, the interface between retrieval and judgement produces errors that cannot be attributed to either stage alone.

Interpretable entailment and structured representations. Silva et al. (2018) construct a knowledge graph from natural-language dictionary definitions so that entailment decisions become auditable through organised semantic relationships. Our Evidence Passport — Claim → Vote → Source → Stakeholder → Position — shares the goal of making evidence relationships inspectable, but is currently a *provenance-oriented diagnostic representation* rather than a verified knowledge graph: the chain records *where* a citation was found, not whether it *establishes* the claim, and the system has not independently adjudicated the semantic sufficiency of those relationships. This gap between source occurrence and semantic support is what structured-representation approaches are designed to make auditable, though it is not established that their architecture would resolve our stakeholder-attribution and evidence-sufficiency problems without independent semantic adjudication.

Sentence embeddings. SentenceSwissBERT is an adaptation of SwissBERT (Vamvas et al., 2023) using a SimCSE-derived procedure (Gao et al., 2021), related to but distinct from Sentence-BERT (Reimers and Gurevych, 2019).

Positional sensitivity. Our C→D comparison relates to Liu et al. (2024) but is a 15-case, single-run, short-context observation, not a replication of their multi-document findings.

Explanation faithfulness. Following Turpin et al. (2023), we flag explanation–label inconsistencies (rows 1344, 964) and do not assume that model-generated explanations faithfully describe the internal decision process.

Selective classification. The coverage collapse is not a calibrated reject-option system (Geifman and El-Yaniv, 2017). No risk–coverage curve was computed; no abstention threshold was calibrated.

Metric selection. Hutiri et al. (2024) caution that metric choice changes conclusions. Our accepted-only accuracy excludes rejected cases; all denominators must remain visible.

Statistical methodology. Paired comparisons use exact McNemar/sign tests following Dietterich (1998).
The official primary metric is three-way held-out Macro-F1. The practical goal is valid NLI output; accepted-only accuracy is insufficient. Neutral rejection remains an unresolved scoring risk. Rejected cases produce `label: null` and a non-zero batch exit; organiser handling is unconfirmed.

5. Limitations and Reproducibility

All measurements are single runs on exposed development data. Denominators differ across panels; x-axis metrics differ; Figure 2 presents only the two genuinely comparable points from the 54-case paired comparison.

The 54-case comparison uses 70B, the selected 70B submission model. Retrieval query, prompt, schema, identity gate and citation mechanism changed simultaneously — no factor can be attributed independently.

The 15-case evidence diagnostic was selected post-hoc after observing failures. Curated evidence was designed with knowledge of claims and gold labels.

Human annotation. No independent human semantic adjudication or inter-annotator agreement is available. A blind protocol exists in the long-form reference, but no operational blind interface or collected annotations exist. Semantic sufficiency remains unverified; error attribution is exploratory.

A first exploratory annotation pass was AI-generated (Qwen-family assistant), a *different model family* from the Apertus predictors but **not independent ground truth**.

Dataset provenance: HackApertus-provided MNLI-over-Swiss-Voting-Booklets v1.1. Twenty-four of 60 dev rows duplicate claim/reference/label triples in the 1,428-row complement. These are exploratory development evidence, not unbiased generalisation estimates.

5.1. Reproducibility Passport

Layer	Status
Tracked source + evidence	`make test` (portable, no model/data needed). `docs/evidence/` supports SHA-256 byte-identity checks; `courage_data.py` verifies provenance when local raw records exist (byte identity ≠ semantic verification).
70B paired development	Requires CSCS API key; not reproducible offline.
MLX 8B diagnostics	Requires Apple Silicon + cached model; not the submission target.
Official 70B submission	External CSCS; authorised credentials required. See release smoke record.
Git-ignored records	`data/experiments/`, `data/eval/` not distributed; `fetch_eval_data.py` downloads the dataset only, not model predictions. Single runs; no variance control.

Reproduction starts with `docs/RUNNING.md`. The diagnostic at `docs/experience/courage-to-be-wrong/` checks provenance hashes when raw records are available; without them, `facts.json` displays static values.

6. Future Work and Conclusion

Four controlled experiments are specified in the long-form reference (§9.1) — oracle-evidence control, evidence-order ablation with repeated controls, `StanceAttributionChallenge` and `BlindAdjudicationProbe` — none has been executed.

VoteLens contributes a failure anatomy for document-grounded NLI over Swiss voting booklets. The headline evidence shows that output-contract validation can destroy coverage without improving correctness: 18 of the candidate's 29 rejected outputs were gold-correct, and 92% conditional accuracy on accepted answers concealed 42.6% strict correctness overall — a system can look reliable when evaluated only on accepted answers while rejecting many otherwise correct ones. The supporting evidence shows that retrieval quality does not straightforwardly transfer to NLI quality and that metric selection determines whether the system appears reliable. None of these findings is a causal result; all require held-out replication with independent semantic adjudication before deployment claims are warranted.

AI Assistance Declaration

Generative AI was used **substantially** in this project: Qwen-based coding agents with OpenCode; ChatGPT for research planning, interpretation and drafting assistance; and Codex for independent cross-model repository auditing and review. Functions included code generation and modification, debugging and test generation, experimental planning, data-analysis assistance, literature discovery, technical report drafting, repository auditing and scientific claim checking. The team directed this work and remains responsible for all submitted content. Automated tests and recorded results were used to check applicable claims. Cross-model AI auditing is not independent human verification, and no independent human semantic adjudication was performed (§5). The full declaration is in the repository's `technical_report.md`; the separate UZH ECTS declaration is a different document.

License

Report: CC-BY-4.0. Team-authored code: Apache-2.0. The public Docker image bundles SentenceSwissBERT encoder weights (CC BY-NC 4.0), but no Apertus LLM weights or booklet PDFs. The team completed its licence review; upstream terms remain applicable (see root notices).

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Model cards: SentenceSwissBERT (`jgrosjean-mathesis/sentence-swissbert`); Apertus v1.5 8B/70B (`swiss-ai/`). Long-form reference: `docs/TECHNICAL_REPORT_LONG.md`.